

AKSHAY PISAL

Junior AI Engineer | Machine Learning | LLMs | Python | LangChain | Generative AI

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Professional Summary

Junior AI Engineer with an MSc in Artificial Intelligence from Brunel University London and 2+ years of industry experience across machine learning, data science, and AI engineering. Currently building AI-powered analytics tools at Untapped Shop using Python, OpenAI API, and LangChain. Background includes hands-on work with deep learning models, NLP, large language models, and data pipelines in both research and commercial settings. Looking for a Junior AI or ML Engineering role in London where I can keep growing and contribute to real-world AI products.

Technical Skills

Programming Languages: Python (Advanced), SQL, Bash

AI and LLMs: OpenAI API, LangChain, LangGraph, Hugging Face (Transformers, PEFT, Datasets), Prompt Engineering, RAG Pipelines, Fine-Tuning

Machine Learning: TensorFlow, PyTorch, Scikit-learn, CNNs, LSTMs, Seq2Seq, Attention Mechanisms, Transfer Learning, Feature Engineering

Data Engineering: Pandas, NumPy, Apache Spark, Oracle Spatial, Large-scale data processing, EDA

Cloud and Tools: AWS, Azure ML, Docker, REST APIs, Git, GitHub, Jupyter Lab, FAISS, AstraDB, Pinecone

Work Experience

Junior AI Engineer

Untapped Shop, London

March 2025 - Present

- Built LangChain and OpenAI API pipelines that pull retail data into a conversational analytics tool, letting non-technical staff ask questions about sales and inventory without needing reports or spreadsheets.
- Developed a GPT-powered analytics prototype that went into live use, cutting out manual reporting and giving the business team direct access to revenue data on demand.
- Tested and refined prompts through regular evaluation cycles, which improved how consistently and accurately the GPT models responded to retail-specific queries.

AI Research Assistant

Brunel University London

August 2024 - November 2024

- Ran supervised learning experiments with TensorFlow and Scikit-learn, testing different CNN and classification model setups to work out which configurations gave the best results.
- Prepared raw research datasets for model training, handling cleaning, feature engineering, and preprocessing to make sure the data was consistent and the model results were reliable.
- Kept detailed records of all experiments, settings, and outcomes in Git, which made it much easier for the team to pick up where previous work left off and repeat results.

Junior Data Scientist

HERE Technologies

August 2021 - August 2023

- Maintained and improved Python and SQL data pipelines that processed large volumes of road network data every day, feeding into HERE's mapping platform used by automotive companies worldwide.
- Used spatial regression models to find and fix errors in geospatial datasets, which improved the accuracy of the map records and reduced the amount of rework needed downstream.
- Automated the data validation and cleaning process in Python and SQL, removing a lot of the manual checking that was slowing the team down and making it easier to handle larger datasets.

Technical Projects

Agentic RAG System with LangGraph and AstraDB

LangGraph | LangChain | AstraDB | Groq | Gemma2-9B | Chroma | Wikipedia | ArXiv

- Built an agentic RAG system using LangGraph that automatically decides whether to pull answers from a vector database or search Wikipedia and ArXiv live, depending on what the question needs.
- Set up the document storage and retrieval side using AstraDB as a cloud vector store with Hugging Face embeddings, so the system can search across a large document set quickly and consistently.

NVIDIA NIM RAG Application

Python | LangChain | NVIDIA NIM | LLaMA 3 70B | FAISS | Hugging Face | Streamlit

- Built a RAG application that loads US Government Census PDFs, splits and embeds the text using FAISS, and lets users ask questions about the documents through a Streamlit interface backed by LLaMA 3 70B via NVIDIA NIM.
- Handled the full document ingestion pipeline from raw PDF to searchable knowledge base, getting retrieval times down to under a second.

Custom Domain-Specific LLM Assistant

Python | Ollama | CodeLlama | Modelfile | Gradio | REST API

- Created a custom AI assistant by writing an Ollama Modelfile that gives CodeLlama a specific system prompt and persona, making it behave like a focused coding tutor without needing to retrain the model.

- Wired up a Gradio chat interface that remembers the conversation history and talks to the model through a local REST API, giving the whole thing a proper chat app feel that runs entirely offline.

Education

Masters in Artificial Intelligence

Brunel University London

September 2023 - September 2024

Bachelor of Computer Science

SIES College of Science, Commerce and Arts

June 2016 - November 2019

Key Achievements

- Won an AI/ML hackathon, building and presenting a working AI solution within the time limit to a panel of industry judges.
- Invited to join the SDG Software Developer Group in London and competed in a hackathon focused on using technology to address UN Sustainable Development Goals.
- Attended the Microsoft London AI Tour and saw live demos and sessions covering Azure AI, Copilot, and enterprise AI products from the Microsoft team.

Certifications

Complete Generative AI Course with LangChain and Hugging Face

Generative AI Engineering and Fine-Tuning Transformers

Generative AI Using Hugging Face